



# Mould in indoor environments: The role of heating, ventilation and fuel poverty. A French perspective

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## ABSTRACT

This paper is based largely on an expert assessment on mould in indoor environments carried out by the French agency for food, and environmental and occupational health and safety (ANSES), within the national Action Plan on Indoor Air Quality. The aim is to provide an overview of the knowledge of indoor mould in France, to determine the factors influencing mould in buildings from an energy perspective, and to investigate the link with regulations dealing with housing construction and refurbishment. The purpose of the assessment is, ultimately, to make recommendations as to how to reduce the presence of mould in indoor environments.

The paper is the result of an analysis of different kinds of data and is not a systematic review of the existing literature on indoor mould. We used data from surveys investigating mould contamination in indoor environments in France and academic publications focusing on energy issues, fuel poverty, building material, and their consequences on mould development. The data also drew on hearings with French stakeholders and an international consultation with national agencies and authorities.

The results show that visible mould is estimated to be present in between 14 and 20% of dwellings in France, a number that is largely due to the fact that old dwellings do not respect thermal and ventilation regulations and that occupants' socio-economic characteristics can lead to fuel poverty or situations of overcrowding. The lack of a specific regulatory framework concerning mould prevention and remediation in France is highlighted by the stakeholders' answers and experience in other countries (in particular the Nordic countries).

The recommendations in the article insist on coherence between thermal regulations and the risk of mould development, development of training for professionals, and control of constructions. Propositions for management strategies concerning assessment of exposure are also presented.

## 1. Introduction

In France, people spend a large part of their day in indoor environments [61] and such environments can lead to important health issues. As the Ottawa Charter for Health Promotion [62] emphasizes, dwellings are an important determinant of health because people are exposed to a large number of indoor pollutants (chemical substances, biological or physical agents) in their homes, with various repercussions in terms of public health [25]. Therefore, housing characteristics represent significant levers of action for public policies aimed at improving the quality of life of individuals through an improvement in the quality of indoor air. In this context, the health impacts and economic consequences of mould deserves special attention.

The Institute of Medicine (IOM) and the World Health Organization (WHO)'s reviews of the scientific evidence for relationships between damp or mouldy indoor environments and the manifestation of adverse health effects highlighted respiratory and allergic symptoms [3].

In recent decades, many studies have shown associations between indoor exposure to mould and respiratory health issues. Meta-analyses and systematic reviews have provided consistent evidence of relationships between mould exposure and asthma [5], and the incidence and exacerbation of asthma and rhinitis [6–8].

Two identified studies estimating costs related to indoor mould focus on the United States or Scandinavia. Mudarri [9] updated the total annual cost to society attributable to dampness and mould: 3.7 (2.3–4.7) billion dollars for allergic rhinitis, \$1.9 (1.1–2.3) billion for acute

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bronchitis, \$15.1 (9.4–20.6) billion for asthma morbidity, and \$1.7 (0.4–4.5) billion for asthma mortality [9].

In 2013, the executive committee of the Finnish Parliament stressed that very substantial savings could be made by giving priority to prevention and remediation in Finland [10]. The costs to society of health outcomes related to moisture damage and mould exposure in Finland are estimated at 450 million euros per year [11], which corresponds to 0.17% of Finland's GDP in 2014, to which the ongoing cost of rehabilitating poorly maintained buildings (ventilation system, repair of water damage, etc.) must be added. In Finland [10], the annual cost to municipalities of rehabilitating public property damaged by moisture and mould amounts to 5 billion euros (1.9% of Finland's GDP in 2013), and health problems caused by severe moisture and mold damage is estimated to 450 million euros a year. In Sweden, the National Board of Housing, Building and Planning (Boverket) estimated in 2010 that the overall cost of having less than 5% of Swedish buildings with moisture-related damage in 2020 would be between 8 and 10 billion euros (based on 2008 price estimates). This cost amounts to 16 to 20% of the country's 2007 GDP.

The problem of mould in buildings concerns different populations (children, the elderly, etc.), depending on individual vulnerability or the socio-economic context. Regardless of the perspective chosen, the energy issue will always be central, as indoor mould development is linked with heating, ventilation and insulation within dwellings and buildings in general.

In France, indoor pollutant levels have been investigated in various environments, including homes, schools, and offices [1,2]. In the framework of the national Action Plan on Indoor Air Quality, the French Agency for Food, and Environmental and Occupational Health and Safety (ANSES) carried out an expert assessment in 2014–2016 dealing with mould in indoor environments. ANSES relied on a working group of 13 experts with complementary areas of expertise (epidemiologists, clinicians, toxicologists, microbiologists, mycologists, sociologists, health geographers, and building engineers) and the ANSES Scientific Committee on assessment of the risks related to air environments.

The aim of the present review is to describe the origin of mould in indoor environments and the determining factors.

The review also focuses on the situation in France, considering mould exposure in indoor environments and French public policy and regulations dealing with housing, construction and refurbishment intended to manage mould exposure. This includes discussion on the implementation with some stakeholders. Looking abroad, interesting examples are provided to highlight recommendations.

## 2. Material and methods

This review is one of the deliverables of a working group of 13 experts, including the authors of the current paper) commissioned by the French Agency for Food, and Environmental and Occupational Health and Safety (ANSES) to give an expert assessment of the exposure to mould in indoor environments. A full report was published in French in August 2016 [11] and a short ANSES opinion is available in English [12]. This work supports the implementation of risk management measures in France.

A literature search was carried out using computerized bibliographic databases (PubMed and Scopus) with different groups of keywords (MESH and free terms), supplemented by the collection of French data from national field surveys and international statements identified through specific consultation.

Starting from the report of the Institute of medicine (IOM) published in 2004, the report of the French Higher Council for Public Health published in 2006 [13], and the World Health Organization report [4] covering publications up to 2007, a systematic review of publications was conducted from 2006 to 2014, completed by references published between 2015 and 2017 contained in personal databases, in the aim of identifying new findings and making specific recommendations

concerning exposure to moulds in indoor environments.

The analysis of literature was conducted in four steps, according to the four-phase flow diagram of the PRISMA statement: identification, screening, eligibility and inclusion.

We chose “mould” and “fungi” as keywords. For Scopus, the search was based on three groups of free terms: 1/mould\*, fung\*; 2/air\*, dust\*, material\*, surfac\*, indoor\*, 3/Humidit\*, temperature, conditions, model, prediction, growth, “air exchange”, “airflow”, “air conditioning”, insulation, ventilation. In the framework of ANSES's expertise, broader searches in Pubmed and Scopus dealing with health effects and measurement of mould in indoor environments were made and completed the records concerning building factors and population. We identified 383 records after excluding duplicates.

After screening on title and abstract, 123 records were selected as follows.

We included records dealing with material vulnerability, mould growth conditions, airflow, mould remediation, population at risk, and intervention studies.

An international consultation took place with national agencies and authorities (in Europe and North America) in the areas of health and/or occupational safety. The aim was to learn about the practices implemented abroad that were associated with exposure to indoor mould, including the applicable regulations, and also to identify the research and development themes in this area (metrology, exposure, toxicology, epidemiology and construction). Another questionnaire was developed and sent to the 26 organizations presented in Table E1 (online supplement). Eighteen responses were collected by email and post, representing 11 countries (response rate of 73%), mainly from northern Europe.

Final inclusion criteria, leading to the review of 45 papers, were specific studies underlying findings concerning mould growth in buildings and populations at risk.

The collection of information also drew on hearings with different stakeholders involved in mould management, at both building and health levels, in order to collect their experience and expectations on the issue of contamination of housing by mould. This approach did not aim to collect all the points of view, an unrealistic objective, but to supplement the literature with feedback from stakeholders in the field.

Three themes were selected to organize the collective hearings in the form of debates bringing together several players involved with each theme (Table 1).

Data were collected from the Regional Health Agencies (ARSs) in connection with their activities, in particular in the framework of the regional environmental health action plans (PRSEs) and their missions to combat unhealthy housing, governed by the French Public Health Code (Articles L.1311–4 and L.1331–22 to L.1331–31). A questionnaire was drawn up and transmitted by email to all regional administrations. Twenty-one responses were collected by email and post, representing 16 regions (response rate of 60%).

The following elements thus come from the literature review, the hearings and the national and international consultation.

## 3. Origin of mould development in buildings from an energy perspective

According to the American Society of Heating, Refrigerating and Air-Conditioning Engineers (ASHRAE, 2012), several parameters that will increase the risk of mould can be identified.

Some of them, mainly due to high level of relative humidity, are linked with building and Heating, Ventilation and Air-Conditioning (HVAC) systems: lack of ventilation, insufficient heating, spread of contaminants through the heating system, type of ventilation system, internal moisture sources, unvented crawl spaces, and damaged building materials.

Others, linked to the presence of water in the building elements, are due to environmental/outdoor conditions (rainwater infiltration,

**Table 1**  
Description of hearings on the issue of contamination of housing by mould.

| Theme                     | Format                          | Date       | Players                                                                                                                                                                                                                                                        |
|---------------------------|---------------------------------|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Health                    | Collective hearing <sup>a</sup> | 03-04-2015 | 4 stakeholders, two from local services (SCHS); one physician from the pneumology department of Paris public hospital (APHP) who was also a member of the French paediatric pneumology and allergology society (SP2A) and one indoor environment adviser (CEI) |
|                           | Individual exchange             | 24-09-2015 | A physician, president of French Medical Mycological Society (SFMM)                                                                                                                                                                                            |
| Administrative procedures | Collective hearing <sup>a</sup> | 16-09-2015 | 3 stakeholders, two from regional or local health administration (ARS) and one from national administration (DIHAL/PNLHI)                                                                                                                                      |
| Work in buildings         | Collective hearing <sup>a</sup> | 16-09-2015 | 5 stakeholders, two from regional or local health administration (ARS), one from the French environment and energy management agency (ADEME), two from consulting companies (Médico, Alhiconseil)                                                              |
|                           | Individual exchange             | 21-09-2015 | a French association for housing improvement (SOLIHA)                                                                                                                                                                                                          |

<sup>a</sup> During a working group meeting.

increased humidity, plumbing leak and flooding). Structural and functional matters can be distinguished that depend on how building structures and systems function or on the characteristics and behaviour of occupants.

The principal causes of development of mould in building elements (see Fig. 2) can be classified as having two main origins (Fig. 1): high level of relative humidity in the air and presence of water in the building element.

Each of these two origins has different causes that are detailed in the tree diagram (Fig. 1). First, water (liquid form) in the building element is mainly due to a vulnerability of the structure that may follow damage or inappropriate management of the building in its design (incorrect layer sequence for water vapour diffusion, which generates condensation in the element), construction (building materials used too wet), or occupation (building occupied too early, drying hampered by occupant behaviour) (Fig. 1). Otherwise, water can enter the building envelope or structure by two routes. On the one hand, infiltrations can be observed related to a structural sealing defect. The main consequence is very often the spread of mould within the construction. On the other hand, humidity can penetrate the building envelope or structure through capillary rise, especially in parts of the building connected to the ground. The porous structure of the material allows the water to progress, essentially in liquid form, depending on the capillary tension (which depends on the pore radius). These topics are not developed in this paper, since they are not directly linked with energy but predominantly with building construction and design.

The second identified parameter is a high level of relative humidity in the air. This parameter is directly linked with energy matters, as humidity largely depends on indoor temperature and ventilation (Fig. 1). Thus in “normally occupied and used” buildings, construction elements with a surface temperature that is too low, for example because of, defective thermal protection, thermal bridges, or the surface being insufficiently heated. In France, the maximum number of occupants of a dwelling is specified during its design. The ventilation rate of outdoor fresh air will be around 20 m<sup>3</sup>/h, corresponding to a normally occupied and used room. Due to fuel poverty, especially in cities, some dwellings are occupied by more people than initially expected. It is observed that these buildings are much more sensitive to mould problems. As indicated, the relative humidity of the ambient air is strongly associated

with the efficiency of the ventilation system or heating. By way of illustration, a ventilation system that is not maintained or is undersized with regard to a volume of air to be treated will not ensure a sufficient air renewal rate, thereby generating an increase in humidity of the ambient air.

In practice, the main reasons for the presence of damp and mould are: lack of ventilation, inefficient heating systems, vulnerable wall materials and building structural problems.

### 3.1. Energy efficiency and mould: ventilation systems and indoor temperature

Several cross-sectional studies have highlighted the link between the presence of mould in dwellings and ventilation. The study by Roussel et al. [27] involved 500 rooms in 128 dwellings in France and sought to qualify the dwellings in which high concentrations of mould were found in the air. Dwellings without a mechanical ventilation system (single flow or double flow) ( $p = 0.003$ ) have a significantly higher concentration of mould.

The role of ventilation in dwellings is to ensure hygiene, comfort and preservation of the structure of the enclosure. Thus, ventilation affects air quality by limiting the accumulation of pollutants and especially water vapour. The effectiveness of the ventilation depends on the external conditions, but it remains a determining factor for the internal conditions in all climates. The absence of ventilation leads to confinement of the living space, which can have consequences for the quality of the indoor environment and the building itself. Mechanical ventilation, which is a way to control and optimize air flow, has been systematic in new buildings in France since the thermal regulation of 2012 (RT2012, 2012).

Even if dwellings with mould are not characterized by a particular type of ventilation compared to other dwellings, the presence of obstacles preventing the availability of sufficient renewable air is very often noted. Case studies highlight that malfunctions in building ventilation systems, as well as occupant behaviour, are frequently associated with moisture and mould problems [28]. In many cases, the way the building is operated by the user is also involved in mould origin. Most of the time, a reduction of the ventilation, unwisely used to decrease heating consumption in winter, increases mould growth.

In order for ventilation to be effective, the energy gains imposed by thermal regulations must not be made to the detriment of air quality. Energy consumption has been mainly related to ventilation (ventilated airflows and power consumption of auxiliaries) in new buildings in France in the last 10 years [29]. In France, dwellings built before the thermal regulation RT2000 generally have heat losses through the envelope (opaque walls and windows), which are much greater than ventilation losses. On the other hand, in recent dwellings, heat losses through ventilation often exceed the losses by the walls, because the new walls and windows have more efficient insulation.

In France, most of the heating systems in dwellings are static heat exchangers (convectors, radiators, heating floors). The ventilation systems are natural ventilation (for buildings before the 1970's), and simple flow mechanical ventilation (mainly by extraction, in contrast to the insufflation systems in other EU countries).

The ventilation rates vary according to the size of the house or dwelling, but it is approximately 20 m<sup>3</sup>/h.occupant. This rate is close to other international standards (ASHRAE 62).

Concerning indoor temperatures, the regulations (RT2012, Work legislation, etc.) advise around 19 °C during winter.

In the case of high relative indoor air humidity, the surface temperature of the building element is very important: if the surface is not sufficiently heated, condensation appears, thus creating suitable conditions for mould development [30]. A classical way to avoid water condensation on and inside the wall under winter conditions is to increase the indoor air temperature. This will avoid cooling of the wall surface, which will be in contact with indoor moist air. So efficient

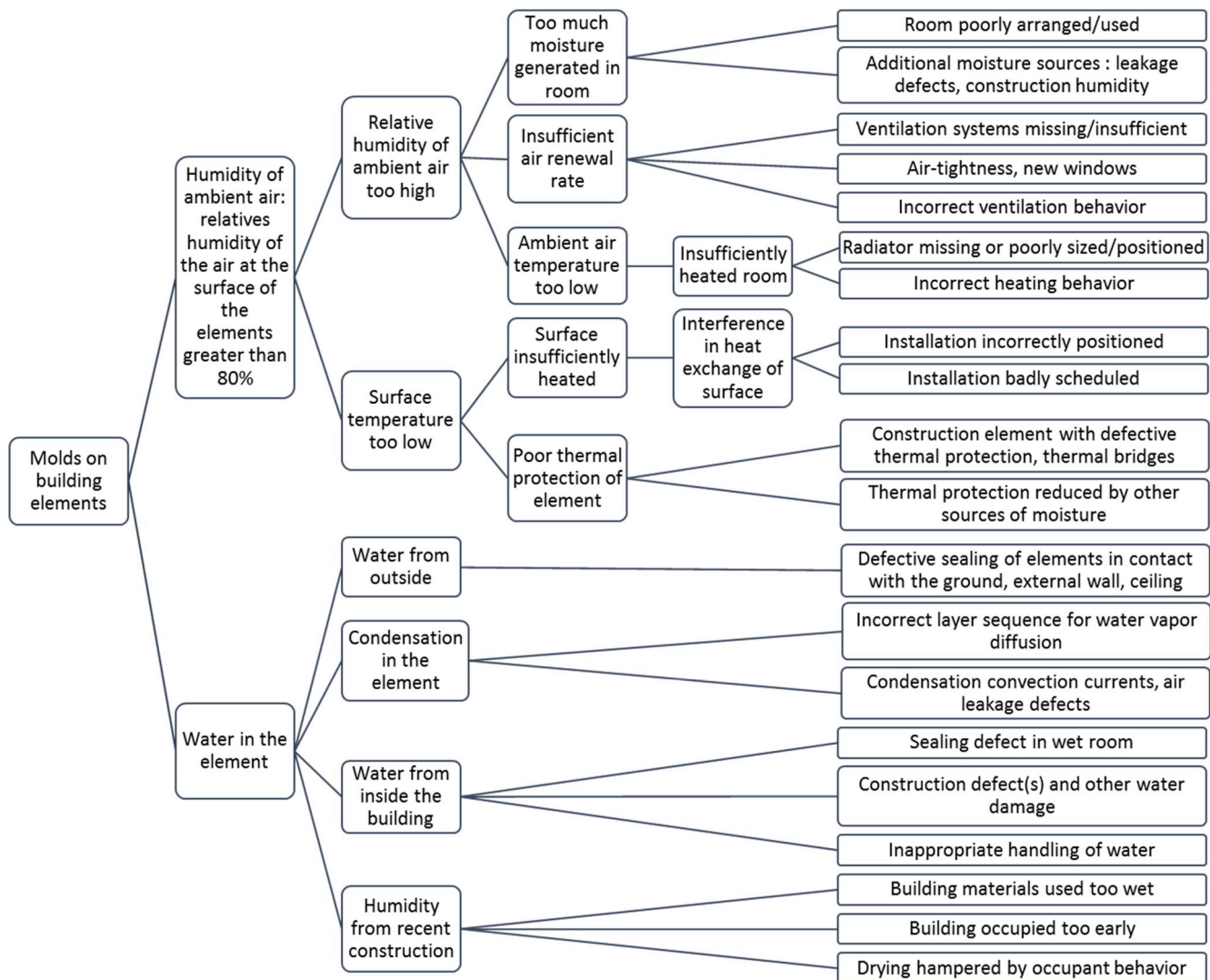


Fig. 1. Causes of mould on building elements (translated from Ref. [26]).



Fig. 2. Mark of mould in a corner corresponding to a thermal bridge or a capillary rise (picture without copyright).

heating systems and insulation materials have to be used.

Usually, indoor air temperature must be maintained at 20 °C to ensure comfort. To decrease the use of energy, some studies recommend adopting a lower indoor air temperature (around 18 °C). It has been

shown that, in recent dwellings, 1 °C less for indoor air temperature causes 8% saving in heating energy [31]. However, this suggestion must be qualified as this decrease of temperature will cause lower surface temperature and will thus increase the risk of mould on the indoor cold walls. Assuming indoor air with a certain specific humidity (kg water vapour/kg dry air), if a dry temperature decrease is observed, it will increase the relative humidity (%), which could lead to condensation, since the moist air will be closer to the saturated state. If the wall temperature is lower than the dew point, condensation will occur (IEA, 1996). So there is a real conflict here between energy savings and mould risk.

Fungal development in dwellings involves complex interactions and it is difficult to quantify the part played by each factor. However, besides ventilation and indoor heating, other significant factors that may contribute to mould development will be addressed in the following sections: building structures, surface materials, insulation, energy poverty, and lifestyle.

### 3.2. Building structures and surface materials, other sources of mould development

According to the IEA [30], if the construction elements and structures are implemented in a way that facilitates the access of water from precipitation, surface water, groundwater, or water of urban networks,



the mould risk is greatly increased. Around building elements located underground (and under-floor-type structures in crawl space or floor), the lack of appropriate solutions will increase water infiltration and cause moisture development. If condensation occurs on the underside of the roof, or where the roof is not sufficiently watertight to prevent the penetration of water, the underlying structure will be contaminated by mould. In floors, walls and ceilings that are expected to be exposed to water spills, water leaks, or recurrent condensation, surface materials will be especially concerned by mould growth.

Thermal bridges are involved in condensation risks, which can lead to mould growth [32].

Finally, bad thermal insulation and single glazed windows can lead to condensation phenomena in or on surfaces in winter conditions. For example, Hulin et al. [33] found the presence of mould to be significantly linked with water infiltration and condensation on windows.

During winter, condensed water can be found in exterior walls, especially when the partial pressure of water vapour exceeds the saturated water vapour pressure [34]. This phenomenon is usually plotted on a Glaser diagram, which brings out the vapour partial pressure inside materials, in steady state conditions. If the excessive level of water in the insulation materials remains for several weeks, it can lead to microorganism development, including bacteria, mould, microalgae, etc. [35]. The proliferation of mould on and inside building materials depends on several factors, which have been demonstrated in recent studies. The main parameters influencing the proliferation of moulds in relation to a support material are as follows [36–39]: chemical nature of the material (especially pH), presence of open porosity - rough surface, use to which the room where the material is present is put, and nature of the mould strain. It should be noted that cellulose-containing materials are suitable media for mould development [40]. The main conclusions are that a strong correlation exists between mould growth and the presence of organic materials [37], that the variability of the hygrothermal properties of a given material (example of several plaster samples) has little influence on mould growth, and that the results for materials containing inorganic binders are quite varied.

Agricultural by-products and other bio-based materials thus represent a potential source of nutrients for fungi and bacteria. However, for the same environmental conditions, different types of bio-based material are not affected in the same way by the growth of moulds [41]. Another material that is conducive to mould development is hemp concrete, as more water is used in hemp concrete than in common concrete. This excess of water, combined with insufficient concrete drying can lead to the development of microorganisms. Moulds can quickly appear on the concrete surface, as the alkali content in the binder is not sufficient to stop the development of mould.

Clearly, a high relative indoor humidity rate can be due to both HVAC weaknesses (unsuitable ventilation, lack of heating) and to construction defects (insufficient insulation, inappropriate materials). Therefore, the development of indoor mould is strongly linked with indoor temperature and ventilation in the dwellings. The residents' behaviour has considerable influence on mould development because such behaviour could cause malfunction in both building systems management and in the use of the dwellings.

### 3.3. Fuel poverty: a major cause of development of indoor mould

Among populations identified as being at risk, those potentially overexposed to mould tended to have lower socio-economic characteristics, leading to fuel poverty or situations of overcrowding.

In metropolitan France, 5.1 million households are affected by fuel poverty, representing 11.5 million people, or 20% of the population, according to the National Energy Efficiency Observatory (ONPE) [44]. The indicators used to estimate the number of French households concerned by fuel poverty combine three approaches:

- the financial effort via a monetary indicator: weight of housing energy bills in the household budget, affecting 2.7 million households;
- restrictive behaviour assessed by means of the difference between actual expenditure and theoretical expenditure (difference between the actual energy bill for heating and the theoretical estimate of expenditure to heat the dwelling normally), affecting 3.4 million households;
- the feeling of thermal discomfort via a subjective indicator, for example the "cold in winter" indicator, affecting 1.3 million households.

These three indicators do not cover the same population groups. By focusing only on households feeling the cold in their homes due to one of the three causes associated with situations of fuel poverty (insufficient heating, poor insulation, financial constraints), ONPE considers that 11.4% of households are concerned. Furthermore, in a study of the general population census data on household energy vulnerability, Cochez et al. [45] showed that the groups standing out were the youngest households (43.1% of households with a reference person under 30), households with an unemployed family member (44.2%), single-person households (33%, compared to 16% of households of couples), working class (26.8% of all working class households) and the elderly (21.1% of retired households). Three studies identified a significant association in French dwellings between thermal comfort and fuel poverty on the one hand and the presence of mould on the other hand [46–48].

In France, the overexposure to mould of households in a precarious economic situation seems mainly linked to fuel poverty, but also too-poor and inadequate housing conditions, characterized by overcrowded dwellings that enhance humidity rates indoors. In 2013, about 8% of French dwellings were concerned by overcrowding,<sup>1</sup> which represented a slight increase, especially in collective housing, whereas, since 1984, overcrowding had been steadily declining. The situation is much worse in urban areas, as shown by Lacroix [51]. Three other studies established a significant association between overcrowding and presence of mould [20,27,33,49].

The situation for the households at risk is even more alarming as mould and moisture in the dwelling is rarely the only sign of environmental degradation. Babès et al. [50] showed that 24% of unemployed people live in a dwelling with a leaking roof, with moisture problems or with visible mould around the windows or doors. According to the analysis of French data for the OQAI's National Housing Campaign described previously, dwellings with mould are more likely to house non-working persons without any particular resources (42% per cent vs. 21% per cent), or persons living with the "minimum integration income" (RMI) (31% versus 9%). More globally, the 2006 Housing survey analysed by Lalande [22] shows that, in France, households with the lowest incomes are the most concerned by humidity in their homes. Thus, 28% of the households with the lowest 10% of income live in a damp dwelling, compared to 12% of those receiving the 10% of highest incomes. Thus the presence of mould in dwellings is a problem that ought to be considered in a global framework characterizing the living environment as a whole.

These findings are consistent with international results and the health effects are particularly important as the most vulnerable individuals are the most exposed to moisture and mould in their homes. Several international studies indicate that housing conditions are one of the mechanisms by which social inequalities can lead to health inequalities [25,52–54]. These health inequalities are often the result of a range of housing-related problems, including the presence of moulds, which in turn are often the result of a precarious household situation and fuel poverty.

<sup>1</sup> The INSEE definition of overcrowding is based on the composition of the household and the number of rooms in the dwelling.

There is clearly an opportunity to target populations of lower socio-economic status in order to prevent indoor exposure to mould. The socio-economic characteristics of the occupants of a dwelling leading to fuel poverty or overcrowding of housing are major problems that should be addressed at the same time as presence of indoor mould and humidity. This is particularly important as the regulation concerning mould in French dwellings is weak.

The review of the literature and the hearings with the professionals working on issues concerning mould exposure also highlight a lack of knowledge about the link between mould exposure and social integration. Living in a dwelling with high presence of mould is, according to international literature [60] and to the professionals interviewed, a noteworthy cause of isolation and depression, for example. Investigations into the consequences of mould exposure for mental health and social isolation, as well as its impact on self-representations and health, will be needed in France.

#### 4. Mould in indoor environments and regulatory framework in France

The previous sections (2 and 3) have shown that the situation is worrying for a rather significant part of French households when it comes to building and housing conditions responsible for mould contamination. As detailed below, indoor mould should be considered an important issue in France. Three French housing surveys that have recently investigated contamination by mould using different assessment methods in various indoor environments show that a significant part of the population lives in housing with mould and, in addition, the national regulations addressing mould mitigation are rather poor.

##### 4.1. Overview of mould in French indoor environments

The presence of visible mould in dwellings in France is estimated at between 14 and 20% based on three French housing surveys (Table 2). These studies are based on nationally representative samples of dwellings. Individuals in the general population completed or answered questionnaires regarding their health status and lifestyle.

Besides the methods used by the national surveys (visual inspection, self-reporting), more specific methods or contamination indicators, such

as the perception of a mouldy odour, the measurement of tracers, counting by culture, are available to characterize the development of mould [16]. There is currently no consensus on the most suitable method but international standards (ISO 16000-16 to 21) specify the requirements in terms of air, surface and material sampling, as well as counting by culture.

Levels of contamination vary according to studies that do not involve the same metrological approaches [17–20].

One explanation of the rather high percentage of dwellings concerned with mould development is the relatively old housing stock in France with only 22% of the dwellings built after 1982. The turnover and refurbishment rates of the dwellings, respectively of 1.1% per year and about 3% per year [21] are slow. Almost one-third of the dwellings were built before 1948, and almost half of those with a bad facade showed signs of humidity in 2006 [22]. This is mainly due to the lack of thermal regulation standards in France before 1974. The proportion of damp dwellings increases with the age of the building, whether it is an individual house or an apartment. Everything else being equal, the risk of damp problems in apartments built before 1948 is almost three times as high as in those built after 1999, and in houses it is almost twice as high [23].

Although there are a few studies assessing humidity and mould in dwellings in metropolitan France, there is very little literature on the situation in the overseas departments, regions and communities (DROM-COM<sup>2</sup>). Compared to what is observed in metropolitan France, in the DROM-COM, climatic and meteorological conditions are different. Materials, types of building construction and housing are also very different. The dwellings are exposed to extreme climatic conditions and, in particular, to high humidity [24]. As a result, dwellings in DROM-COM require specific studies to collect data.

The French situation is rather convergent with the one observed in Europe and the United States, where some studies show that at least 20% of the dwellings have visible mould [3,25]. However, in spite of a rather large proportion of dwellings in France being contaminated with mould, there are few regulations specifically concerning prevention and remediation of indoor mould.

##### 4.2. A weak regulatory framework concerning mould prevention and remediation in France

In France, and more generally in European countries, there are regulations, guide values, action programmes and information strategies concerning indoor air quality. However, according to the WHO Europe review [55] concerning the regulatory context of the European countries, the actions and regulations generally aim at a single theme at a time (ventilation, toxic materials, radon or moulds, etc.), rather than a more global strategy of improvement of indoor air. This is also the strategy adopted in France but, when it comes to mould prevention and remediation, the regulatory framework is particularly weak.

##### 4.2.1. Absence of mould prevention in French thermal regulations

The current French thermal regulation (RT2012) applying to new constructions is the main regulation that deals with energy consumption, indoor air temperature in summer and bioclimatic needs, in new buildings. Thus RT2012 leads to well insulated and energy efficient new buildings, which will most likely reduce mould contamination in general. It is, however, unfortunate that, despite the evident link between thermal conditions and air moisture, humidity and mould risks are not tackled head-on, neither in the current thermal regulation, nor in the future regulation, coming up in 2020 (RT2020) [56]. This absence represents a significant risk of development of indoor humidity and mould in new buildings constructed by the "wet" method, meaning that materials requiring water for their hydration are used *in situ* (concrete,

**Table 2**  
Investigations in French dwellings concerning the presence of mould or moisture traces.

| Source                                                              | Survey                                                                          | Method                                                                               | Results                                                                                                                                                                                                                                              |
|---------------------------------------------------------------------|---------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| National Institute of Statistics and Economic Studies (Insee) [14]. | Housing survey covering 33,000 dwellings, including 7000 in metropolitan France | Visual inspection                                                                    | The presence of signs of humidity on walls is the most frequent flaw in housing (20.65% of housing)                                                                                                                                                  |
| French National Public Health agency [15]                           | Analysis of the 2010 Health Care and Insurance survey (ESPS)                    | Declaration by occupant in questionnaire                                             | 14% of households in metropolitan France live in damp and/or mouldy dwellings                                                                                                                                                                        |
| French Indoor Air Quality Observatory [1]                           | National campaign carried out between 2003 and 2005 in 567 dwellings            | Visual inspection: presence of visible mould and the extent of the contaminated area | 150 rooms of the 4691 inspected (i.e. 3.2% of the rooms) had visible mould. It was distributed over 91 dwellings (16% of dwellings). The surface area contaminated varied from 0.02 to 1.5m <sup>2</sup> per room (median 0.05 m <sup>2</sup> /room) |

<sup>2</sup> Caribbean, Pacific and Indian Ocean regions.

mortars, plasters). It has recently been observed on several projects according to hearings with stakeholders working in buildings (Table 1) that, when the building is “weatherproof and airtight”, mould grows quickly. This is generally the case, as the rooms are not ventilated, because doors/windows are closed to avoid theft on the construction site while workers are away. The hearings with stakeholders working in the construction industry (Table 1) revealed that building professionals are aware of these issues. However, good practices are not always implemented on construction sites. Another source of mould development is linked to water damage during construction, which is also common.

From 2020 onwards, building projects will also need to evaluate the building’s performance in terms of greenhouse gas emissions, achieved via a comparison with the maximum level of greenhouse gas emissions over the building’s life cycle and the level of greenhouse gas emissions caused by products and equipment used in the construction process. This element will act as an incentive for the use of bio-based material as insulation materials, which are many times more sensitive to mould than other materials [39,57,58]. A lower indoor air temperature set point could also be suggested to reduce on-site energy consumption, but would not be consistent with avoiding the risk of mould, as seen in part 4. So, in the French context, future new buildings could be more affected by mould, if unsuitable bio-based materials spread in building practice. The fact that indoor humidity is not directly taken into account in the regulations is clearly a regulatory weakness that works against mould reduction in new constructions.

When it comes to refurbishment of old blocks of flats or houses, various aids and subsidies from the state or local governments help to improve the energy efficiency of a building. In the context of the urban renewal plans launched in recent years by many social housing administrations, buildings dating from the 1950s and 60s have been renovated

thermally.

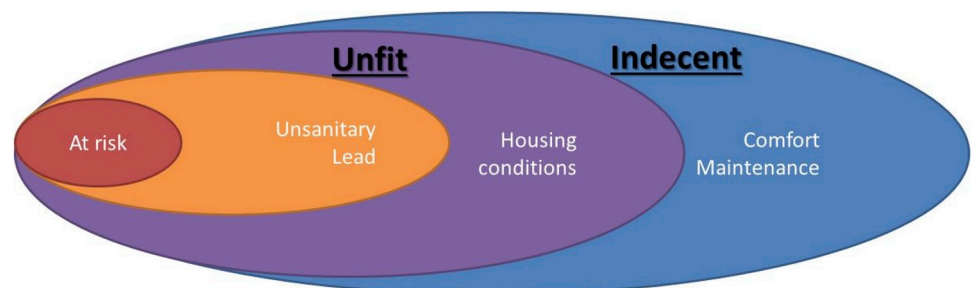
In view of the hearings with stakeholders working in buildings (Table 1), it appears that a lack of coordination among professionals (inadequate ventilation system with regard to the insulation of the housing), create favourable conditions for fungal development. Even if the professionals respect the rules of the art, it sometimes happens that condensation occurs on the inside of opaque walls insulated from the outside. Even when no intervention takes place on the inside of the building (the insulation is often laid on the outside), water can appear on these walls and thus create conditions for mould development.

#### 4.2.2. Housing regulations and mould contamination in France: a general lack of clarity

In France, the regulations are focused on overall management of housing without specific requirements concerning mould contamination. Different notions of housing are defined by the regulations: unsanitary, indecent and unfit. These concepts have been developed with separate objectives involving different managers and stakeholders (Fig. 3).

According to the hearings with stakeholders working in health and housing in France (Table 1), the reporting taken into account by the local authorities is a key issue. They pointed out the complexity of this area, with many stakeholders involved and responsibilities that are often poorly defined, and associated with difficulties in working together and limited exchanges between stakeholders within the same territory.

As there is no French regulation concerning mould assessment in housing, different organizations are authorized to carry out the inspection and different methods are used. In the field, mould contamination is primarily assessed by a visual inspection during the visit to a dwelling but, in some French regions, mould samples can also be taken (for



| Aim              | Safety                                                                                        | Health                                                                                                                                                                          | Conformity                                                        |                                                                                                                                  | Decent                                                                     |
|------------------|-----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| Responsibility   | Mayor<br>Special Policy                                                                       | Representative agent-<br>Special Policy                                                                                                                                         |                                                                   | Mayor<br>General Policy                                                                                                          | Magistrate's court                                                         |
| Regulatory basis | French Construction and<br>Housing Code<br>L511-1 to L511-4<br>L123-3 and L129-1 to<br>L129-7 | French Health Code<br>L1331-25 to L1331- 31<br>L1334-1 to L1334-12 and<br>R1334-1 to L1334- 9                                                                                   | French Health<br>Code<br>L1331-22<br>L1331-23                     | Code for Territorial<br>Communities<br>L2212-2<br>French Health Code<br>(L421-4<br>Règlement sanitaire<br>départemental<br>(RSD) | Law 89-462 and 2000-<br>1208<br>Decree 2002-120                            |
| Stakeholders     | Mayor<br>ADIL (Adviser)                                                                       | Regional or local<br>administration (SCHS,<br>ARS )<br>French family allowance<br>fund (CAF)<br>Mutualité Sociale<br>Agricole (MSA)<br>French National Housing<br>Agency (ANAH) | ADIL (Adviser)<br>local<br>administration<br>(ARS )<br>CAF<br>MSA | Mayor<br>ADIL (Adviser)<br>local administration<br>(ARS )<br>CAF<br>MSA                                                          | ADIL (Adviser)<br>local administration<br>(local commission)<br>CAF<br>MSA |

Fig. 3. Classification of aims of housing regulations.

example using adhesive tape or using bio-impactors in indoor air sampling). The mould assessment can be carried out by an employee of the regional or local administration,<sup>3</sup> an indoor environment adviser (CEI) or a health adviser. Besides the absence of common practices in France, the feedback from the field also indicates that the skills of the professionals responsible for assessing fungal contamination in housing are heterogeneous in the different French regions and, in many cases, insufficient.

## 5. Discussion in the light of international policy

A comparative study of European public policies [59] indicates that the Nordic countries in particular already had an advanced regulatory environment on air and indoor air quality in the early 2000s. In these countries, there is a specific regulation aiming at prevention, reporting and remediation of moisture and mould in indoor environments.

Due to the strong mobilization of the Scandinavian countries through the international consultation, the experiences of Sweden, Finland and Denmark appeared to be of interest for the expert group, helping to identify possible policy strategies for regulations concerning indoor moulds in France.

### 5.1. Actions for prevention of mould

In terms of prevention, the Swedish Planning and Construction Code and Environmental Code incorporates requirements concerning the functions of buildings and materials during construction and the obligation of the owner to maintain these characteristics once construction has been completed. Furthermore, an energy declaration and an inspection of the quality of the indoor environment of housing before sale, lease or renovation and on completion of the construction is compulsory for buildings. There is also compulsory checking of the ventilation system (every 3 or 6 years according to the type of ventilation system) except in houses for one or two families. The information collected during the regular ventilation controls is gathered in a national database allowing most buildings in Sweden to be monitored.

Denmark is the only Nordic country where, in real estate transactions, a certified expert must declare the level of moisture and the presence or absence of mould in the housing. Insurance policies covering property transactions aim to secure the position of the buyer or seller should any damage be discovered that was not apparent during the transaction.

Drawing, in particular, on these examples, on the feedback from the international consultation and on the hearings, ANSES recommendations identified preventive measures to avoid growth of mould in French housing and public buildings as a priority issue.

Firstly, training and information initiatives, among all professionals involved in operations related to buildings (engineers, architects, technical design offices, companies, contracting authorities and prime contractors, artisans, etc.) are necessary to raise awareness about the possible ways to prevent mould development. These professionals need to be encouraged to work in a cross-cutting way with an overall vision of the technical issues related to mould development and remediation, including insulation, ventilation and heating.

Secondly, there is a need to introduce measures into French regulations to prevent the development of indoor mould. The building regulations in France (RT2012, E+/C-label 2020) for new buildings increase the thermal resistance of walls and windows, in order to limit thermal losses. On the other hand, these regulations make controlled mechanical ventilation mandatory, with a very airtight building. During the construction phase (with wet materials) or during the working stage (if an electric power cut occurs, stopping the ventilation), mould development

has frequently been observed. This phenomenon was never observed before, for buildings built before 2012. So, even if new buildings consume less energy (better insulation, controlled renewable air), the mould risk is still significant. In new constructions, a compulsory check on the relative humidity and a search for the existence of thermal bridges by conducting *in situ* measurements over the first winter period of the inhabited phase seems necessary. The expert group also recommends the integration into the next thermal regulation (RT 2020, E+/C-label) of a systematic check on the operation of ventilation systems in the final certificate, along with a mandatory periodic check of the their effectiveness as in Sweden. In France there are energy certificates that are compulsory for real-estate transactions. It would be possible to add to these certificates a mandatory declaration concerning the assessment of the effectiveness of the ventilation and the presence or absence of mould or humidity. Another gainful measure would be a mandatory assessment of the vulnerability of construction materials with regard to mould development before they are placed on the market, on the basis of the existing standards. The labelling of construction materials could then be supplemented to provide information for downstream users. Vigilance is needed to avoid encouraging industrial companies to resort to biocidal products. The expert group also highlights the importance of maintaining the financial aid to encourage work to be carried out to insulate housing.

Thirdly, there is a genuine need to facilitate access of occupants (tenants, owners, etc.) to information on effective measures for preventing mould development in housing, as well as access of the people likely to advise and assist them in this approach. Improved knowledge is likely to encourage citizens and stakeholders to play an active part in the prevention of mould development in housing. For this purpose, the working group has addressed recommendations to the groups concerned. For owners and co-owners, the suggestion was a logbook on the servicing and maintenance of buildings and common areas, if appropriate, in both public and private housing. For occupants, distribution of an information leaflet presenting the rules of use and upkeep when they first arrive in the housing would be useful, whether they have purchased or are renting the dwelling. For local communities, there is a need for technical advice to citizens, in the aim of assessing the thermal and hygro-metric characteristics of the housing at the same time.

More generally, there is a need to strengthen coordination between those involved in the various sectors (construction, energy, etc.) and the authorities and other public bodies in France to improve consistency in risk management related to housing. In particular, the interrelationship between the technical issues associated with buildings, including building methods, insulation, ventilation and heating, needs to be more effectively taken into account by all the stakeholders.

### 5.2. Reporting responsibility

There are also effective mechanisms for reporting mould in a dwelling or building, which are particularly developed in northern Europe. In those countries, public policies are proactive on the issue of damp and mould in indoor environments.

In many countries, such as Sweden and Denmark, reporting an indoor environment problem is primarily the responsibility of tenants and users of the building. There is a standardized procedure in all municipalities. Following a complaint by a tenant or user, the municipality informs the owner of his duty to intervene to treat problem. A certified inspector conducts an assessment of the remedial measures needed and, if necessary, the owner is summoned to deal with the mould. One of the limitations of this approach, where responsibility for alerting belongs to the occupants or users, is that, in some cases, the persons concerned may have difficulty identifying the problems clearly and filling in the complaint form.

Based on the expert assessment, ANSES recommendations aim to improve the reporting procedure, which is not currently normalized in France, regarding the professionals involved and responsible, or

<sup>3</sup> health technician or engineer at the Regional Health Agency (ARS) or municipal hygiene and health service (SCHS).



regarding regional practices. There is an urgent need to establish a single point of contact where all occupants can report fungal contamination in their home, primarily targeting at-risk populations, e.g., through the provision of a standardized reporting sheet as is the case in Sweden. The situation could also be improved by offering training to health and social professionals to raise awareness of their role in identification, reporting and management, especially when the families concerned are over-exposed due to unfavourable socio-economic characteristics.

### 5.3. Intervention and remediation

In most countries, there is no regulation that specifies the remediation procedure for moulds identified in a dwelling or building. In general, the owner and, in some cases, the operator of the premises is responsible for the intervention. Individual houses must be supervised by the owners themselves and are not an administrative matter. In order for remedial work to remove moisture and mould to be properly implemented and effective, technical guidelines on rehabilitation have been drawn up by most countries. This is a necessary but clearly not a sufficient condition for efficient remediation.

Public aids that allow for the renovation or sanitation of a building appear to be a prerequisite for encouraging homeowners to implement remediation works. In Denmark, in case of significant contamination, the owner can request financial assistance for the renovation of dwellings or other buildings, or for demolition. In Sweden, since 2008, tax reductions can be obtained for the repair and maintenance of housing, including remediation of damp and mould problems. Another solution to compel homeowners to renovate homes has been developed by the Netherlands, where a commission (the "Huurcommissie") can intervene in the rental sector by lowering rents if remediation is not executed by the owner. In France, public aid does not exist for mould remediation. According to the professionals consulted who work in refurbishment (Table 1), the incapacity or the reluctance of the owners to invest the necessary financial means is a primary cause of the frequent failures of mould remediation.

A second condition for succeeding in mould remediation, and perhaps even more important than financial aid, is the question of the skills of the professionals involved in the remediation work. According to the feedback that the expert group received from the stakeholders involved with remediation of damp and mould (Table 1), operations very often fail due to a lack of training of the professionals in this field. A report published by the Finnish government [10] shows that, in Finland, despite a regulatory framework that is among the most advanced and strict in Europe, there is an increase of the number of dwellings and other buildings affected by damp and mould. The regulatory and financial measures taken over the past decade have produced no visible improvement in the quality of construction, maintenance and rehabilitation of dwellings and public buildings [11]. The main reasons for the failure of the policies are that the remediation work required and recommended by the government is not carried out, is executed very poorly or is implemented too late. The observed result is a very rapid return of development of mould after the remediation [10].

In the international consultation and the hearings with French professionals concerned by mould exposure, several elements in the Nordic countries' legislations caught the attention of the ANSES expert group. In general terms, the knowledge and skills of the different professionals involved in prevention and remediation of the development of fungal contamination is a key factor for reducing exposure to mould and the associated health effects. Specific skills and knowledge concerning mould are essential for the professionals working in construction, renovation and remediation, as well as for those managing the risks for exposed populations (health professionals and social workers responsible for reporting fungal contamination of buildings, and professionals responsible for assessing it). The establishment of a certification or a professional label for the expert assessment and inspection of renovation work concerning mould in housing, with mandatory national training,

could be a solution to avoid recontamination of dwellings by mould a few months after a professional sanitation intervention. These certified professionals could also make a diagnosis of the housing and propose work with an overall vision of the design elements - mainly by integrating the insulation, ventilation and heating aspects, and also the inspection of new or renovated buildings with regard to mould development. In cases where the remediation is not carried out by a certified company, a verification by an independent expert once the work is finished is essential, according to the professionals interviewed who work in this field (Table 1).

## 6. Conclusions

Indoor mould exposure is clearly a societal issue, as it concerns about 20% of the dwellings in France and in other European and North American countries. Development of mould is largely related to building features: building material, heating, ventilation, insulation, but also to the way they are used by the residents. Indoor mould is particularly present in dwellings where the households are characterized by fuel poverty. The presence of indoor mould is enhanced by the inherent difficulties linked with remediation.

Indoor mould does not necessarily imply a health risk for the occupants, as the risk is linked with both the individual sensitivity and the gravity of the mould contamination. Based on the expert assessment concerning prevention of the development of fungal contamination in buildings, ANSES recommendations focused on preventing the health consequences, especially on the most vulnerable and/or overexposed population groups.

The expert assessment identified several main issues that need to be addressed in order to reduce mould exposure of the general population in France, and particularly of vulnerable and overexposed groups. There is a need for, first, coherence between thermal regulation and risk of mould development; second, development of obligatory information/training of professionals; third, control of construction (materials, insulation, ventilation, etc.); fourth, propositions for management strategies concerning the assessment of exposure with an investigation in the exposed dwelling and management of the fungal risk. Lastly, it is important to emphasize the need for more research to improve knowledge, in particular in the field of human and social sciences with, for instance, new studies seeking to characterize the French situation in terms of consequences of mould exposure in dwellings on mental health and social isolation.

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## Appendix A. Supplementary data

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